Fahad Pathan

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Education & Training

Bachelor of Science (Engineering) in Software Engineering | Shahjalal University of Science and Technology | 02/02/2020 - 21/05/2025 | Sylhet, Bangladesh

Relevant Coursework: Data Structures and Algorithms, Operating Systems and System Programming, Distributed Systems, Software Architecture and Design Patterns, Human Computer Interaction, Database Management Systems, Computer Networks, Object-Oriented Programming, Artificial Intelligence, Machine Learning

Final grade: CGPA: 3.73/4.00 | **Type of credits:** Bangladeshi Credit System | **Number of credits:** 160 (240 ECTS) | **Thesis:** DeepAQNet: A Lightweight Explainable CNN Framework for PM2.5 based AQI Estimation Using Smartphone Imagery in Urban Environments (under review in Heliyon, Elsevier)

Work experience

Software Engineer II | Brain Station 23 PLC | 01/07/2025 - Current | Dhaka, Bangladesh

- Engineered decentralized backends by developing highly scalable Python and Go microservices for distributed environments.
- Designed multi-cloud CI/CD pipelines to automate software deployments across AWS and Azure platforms.
- Developed a RAG pipeline using vector embeddings and Azure AI Search for semantic document retrieval.
- Integrated production AI vision pipelines leveraging AWS Rekognition and Textract for automated data extraction and evaluation.
- Deployed generative AI features into live production systems via AWS Bedrock and Azure AI Foundry frameworks.
- Integrated Stripe payment gateway to enable secure and seamless online transactions within the application.
- Improved the efficiency and throughput of data handling pipelines for high-volume workloads (up to 90% API response time optimized).
- Integrated Mixpanel, Zoho, and Google Analytics to architect a unified data pipeline for monitoring user behavior and engagement metrics.

Email: fahad.pathan@brainstation-23.com | **Website:** https://brainstation-23.com/

Software Engineer I | Brain Station 23 PLC | 01/10/2024 - 30/06/2025 | Dhaka, Bangladesh

- Developed high-performance backend systems using Go and gRPC to facilitate efficient communication between distributed microservices.
- Containerized application environments with Docker to ensure seamless consistency between local and production deployment phases.
- Integrated a business-aware AI chatbot into production to provide automated, context-driven assistance for a growing user base.
- Optimized complex database queries and indexing strategies to significantly reduce search latency.

Research Assistant | Advanced Machine Intelligence Research Lab (AMIRL) | 01/12/2024 - Current | Dhaka, Bangladesh

- Proposed SLA-UNet, a Swin Transformer–CNN hybrid with CBAM attention for skin lesion segmentation; Dice 0.94, IoU 0.88.
- Developed XEndoSIS, a dual-attention (Swin + ECA + SE) model for endoscopic surgical tool segmentation; Dice 0.9492, IoU 0.9036.
- Published a systematic review of ML/DL methods for Alzheimer's disease detection using neuroimaging and clinical data.
- Presented research papers at IEEE conferences (2024, 2025) on medical image segmentation and deep learning.
- Handled data preprocessing, EDA, and visualization to improve model interpretability and evaluation.
- Mentored research interns on end-to-end deep learning projects, from data preparation to evaluation and publication.

Website: https://amirl.org/

Research Intern | Advanced Machine Intelligence Research Lab (AMIRL) | 01/04/2024 - 30/11/2024 | Dhaka, Bangladesh

- Proposed XLTLDisNet, a lightweight CNN (4.6M parameters) for tomato leaf disease classification; 97.24% accuracy (10 classes).
- Co-authored a state-of-the-art review of DL methods for image-based disease classification, detection, and segmentation pipelines.
- Benchmarked pretrained ViT models under transfer learning for multi-class image classification; 99.57% accuracy.
- Developed lightweight models optimized for low-resource environments and deployed as web application using Flask.

Publications

XLTLDisNet: A Novel and Lightweight Approach to Identify Tomato Leaf Diseases with Transparency

A Das, F Pathan, JR Jim, MR Ouishy, MM Kabir, MF Mridha (2025). Elsevier Heliyon (Q1, IF 3.6) Volume 11, Issue 4.

Proposed a 4.6M-parameter efficient CNN for 10-class image based tomato leaf disease classification on PlantVillage dataset. Achieved 97.24% accuracy. Integrated Grad-CAM and LIME to produce visual explanations. Deployed on vercel.

Deep learning-based classification, detection, and segmentation of tomato leaf diseases

A Das, F Pathan, JR Jim, MM Kabir, MF Mridha (2025). Elsevier Artificial Intelligence in Agriculture (Q1, IF 12.4) Volume 15, Issue 2.

Conducted a comprehensive survey on DL-based methods for image-based tomato leaf disease diagnosis. Analyzed classification, detection, and segmentation architectures, preprocessing pipelines, datasets, limitations, and future research direction.

Advances in Alzheimer's disease diagnosis with machine learning and deep learning techniques

T Alam, J Suchi, A Saha, F Pathan, A Das, MM Kabir, MF Mridha (2026). Springer Netherlands Artificial Intelligence Review (Q1, IF 13.9) Volume 59, article number 90.

Systematic review of ML/DL for Alzheimer's diagnosis. Analyzed multimodal neuroimaging datasets, preprocessing strategies, and model performance. Evaluated state-of-the-art architectures and outlined directions for clinically translatable AI.

XEndoSIS : Explainable Endoscopic Surgical Tool Segmentation via Dual Attention Mechanism

F Pathan, J Mohmmad, F Tazrin, A Das, MF Mridha, J Shin (2025). IEEE 28th International Conference on Computer and Information Technology (ICCIT)

Presented a hybrid segmentation model with a Swin Transformer backbone and ECA and SE attention for surgical instrument localization. Achieved 0.9492 Dice and 0.9036 IoU on the Kvasir-Instrument benchmark.

DeepAQNet: A Lightweight Explainable CNN Framework for AQI Estimation Using Smartphone Imagery

F Pathan, MS Hafiz, A Habib. Under review

Developed a lightweight CNN for predicting PM2.5-based AQI from smartphone-captured images across Bangladesh, India, and Nepal (RMSE 25.51 and R2 0.96).

Projects

EcoSync - Smart Waste Management Platform

Developed a comprehensive web app for Dhaka North City Corporation (DNCC) waste management, featuring role-based access, vehicle & facility management, billing, route & fleet optimization, and dashboard statistics. [Github]

SUST Football Tournament Management System

Built a backend service in Go to handle team registration, match scheduling, and live score updates for university football tournaments. [Github]

Vehicle Detection and Tracking System

Developed an automated system for vehicle detection and multi-object tracking using theYOLO11 architecture. [Github] [Deployment]

Skills

Languages | Python | Go | Java | TypeScript | JavaScript | C++ | C

Libraries and Frameworks | ExpressJS | NestJS | Django | FastAPI | Tensorflow | Keras | OpenCV | NumPy | Pandas

Development Tools | Git | Github | Linux | MySQL | PostgreSQL | MongoDB | Docker | AWS | Azure | Redis

Honours and Awards

AI Olympiad - 1st Runner-up (Tertiary category) | Daffodil International University | 03/05/2025

Awarded 1st Runner-up in the National AI Olympiad. Demonstrated a custom Deep Learning model for real-time AQI prediction using smartphone-captured imagery, focusing on architectural efficiency and urban deployment. [Drive] [News article]

University Innovation Hub Program - Winner | Bangladesh Hi-Tech Park Authority (BHTPA) | 04/05/2024

Awarded Winner at the University Innovation Hub Program for co-founding InterviewMate, an AI platform utilizing BanglaBERT and transformers to provide bilingual interview training. Engineered the NLP-driven feedback system to bridge communication gaps for students. Secured pre-seed funding for its technical feasibility and social impact. [Drive] [Program article]

Certifications

AWS Certified Solutions Architect – Associate | Amazon Web Services (AWS) | 31/08/2025

An internationally recognized certification validating expertise in designing scalable, secure cloud architectures, and deployment. [Certificate]

AWS Certified Cloud Practitioner | Amazon Web Services (AWS) | 26/05/2025

An internationally recognized certification validating foundational knowledge of AWS Cloud infrastructure, security, and core services. [Certificate]

Language Skills

English

IELTS (Overall: 8.0, Listening: 8.5, Reading: 8.5, Writing: 7.5, Speaking: 6.5)